

Supplementary Material: Independent Content and Style Evidence for UAV–Satellite Retrieval

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Local working draft: completed-result integration; remaining experiments and final review are pending.

I. COMPLETED SEMANTIC-MODEL EVIDENCE

The official evaluation contains 42 runs and 231 run–task records. The 36 main fits produce 66 task–variant three-seed summaries over 198 run–task records. The six additional fits are seed-1 sensitivity settings and produce 33 further task records. Tables I and II retain all sensitivity results and the matching main Full seed-1 reference for each of the 11 tasks. No SD is assigned to a single fit. The three-seed main and transfer tables in the main paper report sample SD with denominator $n - 1 = 2$; they do not report standard errors, confidence intervals, or significance decisions.

A. Corruption Metrics and Evidence Path

The four dataset–variant seed-1 runs evaluate six query-corruption families at five severities, yielding 660 task-level corrupted evaluations and 22 clean references. The aggregate retains six metrics per corrupted task evaluation (3,960 rows). The R@1 and official mAP plot sources contain 1,320 of these rows, with all tasks and severities retained. Relative retention is $100m_{\text{corrupt}}/m_{\text{clean}}$; the percentage-point drop is $100(m_{\text{clean}} - m_{\text{corrupt}})$ for metric fractions. Neither substitutes for the absolute corrupted metric. All supplied clean denominators are positive; a zero denominator would make retention undefined.

Full uses cached CLIP evidence for clean evaluation and online CLIP evidence for corrupted queries. The 64-image clean diagnostic has no numerical equivalence threshold, so it establishes neither bitwise parity nor negligible effects on ranking. The curves combine pixel corruption with this evidence-path difference. They are seed-1 descriptive measurements, with no pooling across tasks, directions, heights or severities.

B. Clean-Query Diagnostic Definitions

T4’s shared Visual-margin quartiles use linear 25th, 50th and 75th percentile cutpoints, with equality assigned to the lower interval. Full is aligned to Visual query order before applying the same mask. Quartiles are ordered categories rather than equal numerical margin intervals or necessarily equal-sized groups. The 132 margin strata comprise 84 groups meeting the registered 100-query threshold and 48 groups of 20 queries that do not. Eligibility is a count threshold, not evidence of significance. Other entropy and semantic strata remain outside the independently recounted membership/value scope.

For T5, let N be the full task’s query count and let $k = \lceil cN \rceil$ at requested coverage c . Realized coverage is k/N . Each variant independently selects the leading k queries after stable descending ordering by raw Top-1-minus-Top-2 cosine margin. The selective risk denominator is k ; coverage-constrained success uses N and counts queries that are both selected and correct. The selected-set overlap counts membership selected by both variants, not correctness shared by both. At full coverage, success equals ordinary R@1.

The 66 native task–seed–variant series retain ten requested coverages from 0.1 to 1.0 (660 stored points). Saved AURC is computed upstream over all query prefixes, not by integrating those ten plotted points. The 132 paired records use the four requested coverages 0.5, 0.75, 0.9 and 1.0; the 0.75 records are separate observations, not interpolation from the native ten-point display.

The fixed reliability score is $\text{clip}((s_1 - s_2)/2, 0, 1)$, where $s_1 - s_2$ is the cosine margin. It is not a posterior probability. All 15 bins for each of the 66 series are retained (990 bin records): 140 occupied and 850 empty. Empty-bin count is zero while mean score and accuracy are undefined, not zero. ECE is the saved count-weighted absolute score–accuracy gap, a fraction rather than an accuracy percentage. A lower ECE can accompany lower accuracy and is not by itself evidence of improved probability calibration. Visual and Full can place different queries in the same numbered bin.

C. Statistical and Provenance Scope

The 33-comparison family for official Visual-versus-Full results is distinct from the T4 eligible 284-comparison family and the 132 T5 paired records. The completed query audit independently recounted the 132 shared margin strata and recomputed the 66 T5 native and 132 paired query records from their stored per-query inputs. Entropy/semantic-stratum memberships and values were not independently recomputed. T4 exact McNemar calculations were checked from saved discordance counts, with query recount restricted to the shared margin strata. Retained bootstrap intervals were not independently resampled or accepted as new inferential evidence; this draft does not use them to claim significance for the semantic results. These limits do not redefine the separate historical public-baseline analyses.

This result integration formats existing measured summaries. Checkpoint, cache and full image-content SHA evidence is inherited, with documented historical metadata-chain gaps. It introduces no fresh model inference, full-ranking reconstruction or recomputation of AP from all positive ranks. The source tables and detailed provenance accompany this

TABLE I

SEED-1 FULL SENSITIVITY RESULTS: R@1 (%). THE FIRST COLUMN IS THE EXISTING MAIN FULL SEED-1 REFERENCE, NOT A THREE-SEED MEAN. EACH OTHER SETTING USES ONE FIT PER DATASET. NO SD OR SIGNIFICANCE IS INFERRED. VALUES ARE ROUNDED TO THREE DECIMALS; A POSITIVE VALUE THAT WOULD ROUND TO ZERO IS SHOWN AS < 0.001. AN ACTUAL ZERO REMAINS 0.000.

Task	R18/512 reference	R50/512	R18/256	R18/1024
University D2S	31.906	35.890	31.771	32.918
University S2D	45.934	49.929	43.224	46.505
University Street2S	0.931	0.698	0.931	0.969
SUES U150→S	28.700	31.550	26.125	29.800
SUES S→U150	47.500	45.000	38.750	50.000
SUES U200→S	33.150	36.975	32.125	34.925
SUES S→U200	61.250	50.000	46.250	43.750
SUES U250→S	35.975	42.200	35.200	35.700
SUES S→U250	60.000	53.750	47.500	50.000
SUES U300→S	38.775	46.675	36.000	36.200
SUES S→U300	60.000	57.500	60.000	55.000

TABLE II

SEED-1 FULL SENSITIVITY RESULTS: OFFICIAL TRAPEZOIDAL MAP (%). THE FIRST COLUMN IS THE EXISTING MAIN FULL SEED-1 REFERENCE, NOT A THREE-SEED MEAN. EACH OTHER SETTING USES ONE FIT PER DATASET. NO SD OR SIGNIFICANCE IS INFERRED. VALUES ARE ROUNDED TO THREE DECIMALS; A POSITIVE VALUE THAT WOULD ROUND TO ZERO IS SHOWN AS < 0.001. AN ACTUAL ZERO REMAINS 0.000.

Task	R18/512 reference	R50/512	R18/256	R18/1024
University D2S	38.142	41.974	37.913	38.908
University S2D	29.240	34.499	29.009	30.088
University Street2S	2.185	1.969	2.335	2.237
SUES U150→S	37.158	38.837	34.800	38.430
SUES S→U150	34.282	32.035	29.901	32.113
SUES U200→S	41.465	44.323	40.568	43.465
SUES S→U200	40.246	37.113	35.711	36.900
SUES U250→S	44.434	49.131	43.424	44.457
SUES S→U250	43.996	41.645	38.474	39.406
SUES U300→S	47.108	53.093	44.307	45.021
SUES S→U300	45.247	44.647	41.942	40.986

local working draft; these limits apply to both main and supplementary semantic results.

II. HISTORICAL PUBLIC-BASELINE QUERY ALIGNMENT

The analysis uses retained per-query University-1652 outputs from four fully trained public baselines: Global CNN with Google imagery, LPN without Google imagery, LPN, and LPN with CA-HRS. Comparisons align query paths and location labels across methods and retrieval rules. This produces 37,855 drone-to-satellite (D2S) queries nested within 701 location identities and 701 satellite-to-drone (S2D) queries, one per identity. The analysis operates on the stored feature matrices and per-query retrieval outputs.

III. NUMERICAL RESULTS

Tables III–VII provide the numerical values behind the separate historical public-baseline result summaries in the main paper. The exact baseline table uses the public per-query CUDA operation and reverse NumPy sorting. The paired table uses the same exact path-aligned per-query outputs for point estimates, intervals, and tests. The controlled ranking-rule, gallery-scale, and fusion analyses use a separate vectorized ranking implementation; near-tie ordering can differ from exact per-query ranking.

IV. PUBLISHED REFERENCE COMPARISONS

Table VIII lists MEAN and CDM-Net results on SUES-200 at each flight height. The rows follow the image inputs and training settings in their source papers. Table IX reports the University-1652-to-SUES-200 transfer setting from InfoGeo v5, including its relational-distillation variant. These tables retain the published metrics and source table identifiers.

V. RETRIEVAL-RULE SENSITIVITY

Each of four alternatives—CSLS-k10, Reciprocal-k10, AQE-k10, and AQE+CSLS-k10—is compared with cosine ranking for every method and direction. The resulting 32 paired Top-1 comparisons form one complete family. Exact two-sided McNemar probabilities are adjusted with the Holm procedure across that family. Average-precision changes are paired descriptive means; they do not receive binary McNemar inference.

Figure 1 shows that CSLS-k10 increases Recall@1 in all eight method-direction comparisons by 0.97–3.85 percentage points, and all eight changes remain significant after correction. Reciprocal-k10 reduces D2S Recall@1 by 0.18–0.39 points for all four methods; these four reductions also remain significant. AQE-based variants leave Top-1 outcomes

TABLE III

EXACT LOCAL REPRODUCTION OF FOUR FULLY TRAINED UNIVERSITY-1652 BASELINES. SCORES USE THE PUBLIC PER-QUERY CUDA OPERATION GALLERY @ QUERY AND THE OFFICIAL TRAPEZOIDAL MAP DEFINITION. ALL METRICS ARE PERCENTAGES.

Direction	Method	R@1	R@5	R@10	mAP	MRR
D2S	Global CNN + Google	61.91	81.92	87.35	66.42	70.93
D2S	LPN w/o Google	76.62	90.37	93.60	79.72	82.82
D2S	LPN	76.75	90.54	93.97	79.88	83.02
D2S	LPN + CA-HRS	76.94	90.52	93.82	80.04	83.13
S2D	Global CNN + Google	79.60	85.59	87.87	62.73	82.39
S2D	LPN w/o Google	87.02	90.73	92.58	76.79	89.01
S2D	LPN	86.73	91.30	93.30	76.42	88.97
S2D	LPN + CA-HRS	87.45	91.87	93.15	76.58	89.51

TABLE IV

EXACT QUERY-PAIRED R@1 COMPARISONS AGAINST LPN+CA-HRS. CONFIDENCE INTERVALS USE 10,000 PAIRED BOOTSTRAP SAMPLES; TWO-SIDED EXACT MCNEMAR TESTS ARE HOLM-CORRECTED OVER ALL 12 DIRECTIONAL METHOD PAIRS.

Direction	Method A	Method B	$\Delta R@1$	95% CI	p_{Holm}
D2S	Global CNN + Google	LPN + CA-HRS	+15.03	[+14.57, +15.49]	$< 10^{-300}$
D2S	LPN w/o Google	LPN + CA-HRS	+0.32	[-0.04, +0.69]	
D2S	LPN	LPN + CA-HRS	+0.19	[-0.16, +0.53]	
S2D	Global CNN + Google	LPN + CA-HRS	+7.85	[+5.28, +10.41]	2.86×10^{-8}
S2D	LPN w/o Google	LPN + CA-HRS	+0.43	[-1.43, +2.43]	
S2D	LPN	LPN + CA-HRS	+0.71	[-1.57, +3.00]	

TABLE V

CONTROLLED RETRIEVAL RULES AVERAGED OVER THE FOUR FULLY TRAINED BASELINES.

Retrieval rule	D→S R@1	D→S mAP	D→S MRR	S→D R@1	S→D mAP	S→D MRR
Cosine	73.06	76.52	79.98	85.20	73.13	87.47
CSLS-k10	74.68	78.34	82.00	88.80	78.05	90.57
Reciprocal-k10	72.77	76.29	79.82	85.20	73.14	87.48
AQE-k10	73.05	75.68	78.30	85.20	81.16	85.81
AQE+CSLS-k10	73.06	75.51	77.97	85.16	82.60	85.86

TABLE VI

GALLERY-SIZE SCALING FOR COSINE RETRIEVAL, AVERAGED OVER FOUR FORMAL BASELINES. EVERY REDUCED GALLERY RETAINS ALL POSITIVES AND USES THE SAME PATH-HASH-ORDERED NEGATIVES FOR EVERY METHOD.

D2S gallery	R@1	mAP	S2D gallery	R@1	mAP
100	88.03	90.01	1,000	94.76	94.87
250	82.64	85.24	5,000	91.19	88.35
500	77.96	81.02	10,000	89.19	84.42
750	75.31	78.57	25,000	87.05	78.44
951 (all)	73.06	76.52	51,355 (all)	85.20	73.13

TABLE VII

STRONGEST PATH-ALIGNED Z-SCORE FUSION VERSUS THE STRONGEST SINGLE BASELINE (LPN+CA-HRS). FUSION COMBINES LPN WITHOUT GOOGLE, LPN, AND LPN+CA-HRS. INTERVALS HERE USE 10,000 ORDINARY QUERY-PAIRED BOOTSTRAP RESAMPLES; THE MAIN-TEXT COMPLEMENTARITY ANALYSIS REPORTS THE CORRESPONDING LOCATION-CLUSTER SENSITIVITY ANALYSIS. EXACT MCNEMAR TESTS ARE HOLM-ADJUSTED OVER THE TWO DIRECTIONS.

Direction	Single R@1	Fusion R@1	$\Delta R@1$ [95% CI]	Single mAP	Fusion mAP	ΔmAP	p_{Holm}
D2S	76.94	80.85	+3.91 [3.62, 4.19]	80.04	83.52	+3.48	2.71×10^{-163}
S2D	87.45	89.16	+1.71 [0.14, 3.28]	76.58	80.46	+3.88	

mostly unchanged while altering rank-sensitive average precision, especially for S2D. Each comparison holds the trained checkpoint and query set fixed.

VI. QUALITATIVE SELECTION RULE

The Top-5 image plate uses the lexicographically first path-aligned D2S query for which LPN+CA-HRS is incorrect at

TABLE VIII
AUTHOR-REPORTED SUES-200 RESULTS (%) BY FLIGHT HEIGHT.

Method	Direction	150 m		200 m		250 m		300 m	
		R@1	AP	R@1	AP	R@1	AP	R@1	AP
MEAN [1]	UAV → sat.	95.50	96.46	98.38	98.72	98.95	99.17	99.52	99.63
MEAN [1]	Sat. → UAV	97.50	94.75	100.00	97.09	100.00	98.28	100.00	99.21
CDM-Net [2]	UAV → sat.	93.78	95.16	97.62	98.16	98.28	98.69	99.20	99.31
CDM-Net [2]	Sat. → UAV	95.25	92.24	98.50	96.40	99.00	97.60	99.00	98.01

MEAN uses RGB images at 384 pixels; CDM-Net uses generated descriptions and synthesized views at 512 pixels. Sources: MEAN Tables III/IV and CDM-Net Table IV.

TABLE IX
PUBLISHED UNIVERSITY-1652-TO-SUES-200 TRANSFER RESULTS (%).

Method	Direction	150 m		200 m		250 m		300 m	
		R@1	AP	R@1	AP	R@1	AP	R@1	AP
InfoGeo [3]	UAV → sat.	90.87	92.53	94.20	95.12	96.08	96.30	96.31	96.75
InfoGeo (+RD) [3]	UAV → sat.	91.80	93.18	95.40	96.15	96.58	97.08	96.48	97.00

Source: InfoGeo v5, Table 9. Training source: University-1652; evaluation target: SUES-200. RD denotes relational distillation. Input size is 448 pixels.

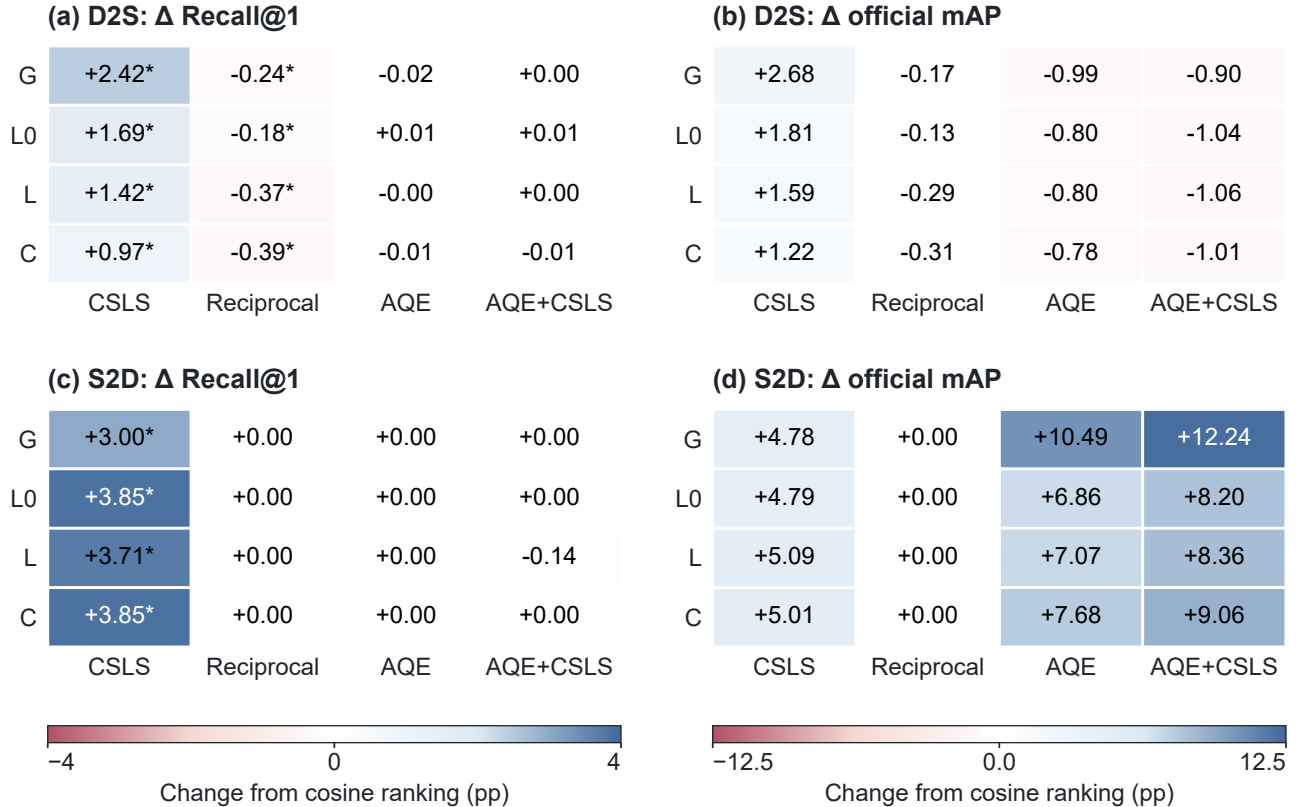


Fig. 1. Controlled retrieval-rule sensitivity on the same path-aligned full-gallery queries. The upper and lower rows show D2S and S2D, respectively; panels (a,c) show changes in Recall@1, and panels (b,d) show changes in official trapezoidal mAP. Values are percentage-point changes from cosine ranking, with $k = 10$ for all rules. Each metric shares one zero-centered color scale across directions. Asterisks denote exact two-sided McNemar tests with Holm-adjusted $p < 0.05$ across 32 comparisons; mAP changes are descriptive. G: Global CNN+Google; L0: LPN without Google; L: LPN; C: LPN+CA-HRS.

rank 1 and the retained three-model fusion is correct. This rule is evaluated before image display. The source table records the

query path, identity, retrieved paths, ranks, correctness, and scores for both rows. The aggregate rescue and harm counts

summarize how frequently this behavior occurs in the query set.

VII. STATISTICAL ANALYSIS AND SUBSET SELECTION

The error-complementarity oracle counts a query as correct when at least one of the four baselines retrieves the correct identity at rank 1. It therefore measures the observed union of their successes. The retained fusion subset was selected on the evaluation set. Its identity-cluster bootstrap intervals condition on that selection and measure variation across re-sampled locations. The broader D2S baseline and retrieval-rule comparisons use paired query images as inferential units. Repeated images from the same location can be correlated, which narrows query-level intervals relative to an analysis that retains location clusters. The machine-readable records contain the resampling settings, comparison families, query counts, source hashes, and software versions.

VIII. SEMANTIC EVIDENCE CACHE AND REPRODUCIBILITY

Table X reports evidence-cache integrity. Each image has one unique relative-path entry. Probability rows are finite, nonnegative within numerical tolerance, and normalized to sum to one. The model revision, vocabulary order, metadata hash, and cache hash are recorded alongside the arrays.

Each run stores command and environment metadata, pretrained-weight records, evidence/data/code hashes, the sampler schedule, training history, checkpoint, and status. Official evaluation additionally stores aggregate metrics and per-query paths, labels, AP, reciprocal rank, margins, Top-1 correctness, and retrieved paths. The checkpoint record identifies its training epoch and associated evaluation outputs.

Reproducing a comparison requires the same text order, frozen encoder, cache arrays, and per-run configuration. These identifiers accompany the probability inputs and retrieval checkpoints.

IX. SEMANTIC-MODEL PROTOCOL DETAILS

The completed main protocol comprises two datasets, six input configurations, and three seeds, for 36 independent 80-epoch fits. Six additional seed-1 full-model fits change ResNet-18 to ResNet-50 or the descriptor width from 512 to 256 or 1024 on each dataset. The completed matrix contains 42 fits. QDFL, FSRA, SDPL, CCR, and MCCG define pending external visual-comparison targets; no completed results for those comparisons are asserted in this working draft.

A. Data Inventory and Training Schedule

The University-1652 evidence inventory contains 146,520 images. The formal protocol uses 137,576 supported official rows and excludes 8,944 auxiliary Google-view images. The SUES-200 inventory contains all 40,200 images. Image paths join the cached evidence to the official task manifest; CLIP scores the pixels of each image against the fixed vocabularies.

Every official training query is scheduled at least once per epoch. A fixed least-loaded-bin allocation gives 652 optimizer

steps per University-1652 epoch and 375 per SUES-200 epoch. Training uses the official training identities and a predetermined final epoch. A separate evaluation command applies the final checkpoint to the official retrieval tasks.

B. Statistical Implementation

For discordant Top-1 counts b and c , the exact two-sided McNemar probability is

$$p = \min \left\{ 1, 2 \sum_{k=0}^{\min(b,c)} \binom{b+c}{k} 2^{-(b+c)} \right\}. \quad (1)$$

The sum is evaluated in log space, and very small values are reported as numerical bounds rather than zero. Holm correction is applied within the declared comparison family. The official semantic-model protocol defines a 33-test family across three seeds and 11 official tasks. It is distinct from the T4/T5 diagnostic records described above; the retained semantic bootstrap intervals have not been independently resampled. The public-baseline analysis uses a 12-comparison exact-reproduction family and a 32-comparison retrieval-rule family. Each adjustment is applied within its full declared family.

C. Experiment Records

The experiment record links the final checkpoint to its training configuration, software and code identifiers, logs, timing records, and path-aligned per-query arrays. The corruption protocol links visual/full seed-1 checkpoints to the corresponding clean-reference evaluation so that each perturbation is measured against the same model.

TABLE X
FROZEN EVIDENCE-CACHE INTEGRITY. HASHES SHOW THE FIRST 10 HEXADECIMAL CHARACTERS OF EACH SHA-256 DIGEST.

Dataset	Rows	Coverage	Cache hash
University-1652	146,520	100%	8a2333d58d
SUES-200	40,200	100%	6c3a82fdcf

REFERENCES

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